

Enhancing real-time urban drainage network modeling through Crossformer algorithm and online continual learning

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ABSTRACT

This study presents an innovative approach to real-time modeling of urban drainage networks, leveraging a highly accurate coupled one- and two-dimensional hydrodynamic model to generate a training dataset for node water levels. By employing global states inferred from monitoring points as model inputs, this study overcomes the limitations imposed by the scarcity of monitoring data and the challenge of capturing all node levels. The Crossformer algorithm, which simultaneously accounts for correlation at both temporal and feature scales, is applied to enhance the precision of simultaneous water level predictions across the network. Comparative analysis of different prediction patterns reveals that extending predictions based on a high-accuracy infrastructure offers more benefits than direct modifications to the algorithm's structure. In addition, this paper pioneered the application of online continuous learning concepts to update the prediction model in real time, achieving a balanced integration of measured and simulated data. Consequently, this paper establishes a complete monitoring-predicting-updating real-time simulation system for urban drainage networks.

1. Introduction

Global cities are confronting the formidable task of maintaining the efficiency, safety, and sustainability of their urban drainage systems amid swift urbanization and changing climate impacts. Essential to these systems' success are precise monitoring, management, and predictive analysis, ensuring operational effectiveness and prompt response (Palmitessa et al., 2022). Crucial to this endeavor is developing models that accurately mirror or predict the drainage networks' hydrodynamic behavior, enabling proactive control and optimization of these systems to address a myriad of urban challenges.

Numerous studies leverage high-fidelity numerical models based on physical mechanisms to analyze the twin primary processes of surface runoff and subterranean sewer convergence. The development of hydrodynamic models has evolved from simple runoff estimates using conceptual models to advanced two-dimensional (2D) overland flow analyses (Chang et al., 2015; Teng et al., 2017). Subsequently, the interaction between sewer networks and overland surfaces has progressed from simple one-way to sophisticated two-way coupling mechanisms (Dong et al., 2021; Hsu et al., 2002; Seyoum et al., 2012). The method currently acknowledged for its highest accuracy is the bidirectional coupling of a 2D surface model predicated on the Shallow Water

Equations (SWEs), and a one-dimensional (1D) network model founded on the Saint-Venant equations.

Building physically based numerical models requires extensive, often hard-to-access data (Rahman et al., 2002). These models are restricted by computational speed and are capable only of calculating the corresponding response under designated input conditions, lacking the foresight to predict future events (Chiang et al., 2010). With artificial intelligence advancements, some researchers have turned to data-driven methods for predictive modeling (Fu et al., 2022; Garzón et al., 2022; Piadeh et al., 2023). Due to budget limitations for sensor devices, research endeavors that leverage historical monitoring data for their datasets are mostly confined to variables within the hydrological sphere, such as runoff quantity and river level or quality (Chen et al., 2020; Kratzert et al., 2018; Xiang et al., 2020). In the context of drainage networks, the scarcity of nodes equipped with monitoring apparatus restricts the observational scope, preventing a comprehensive assessment of the network's condition (Jiang et al., 2021, 2024). To overcome this, some studies have resorted to constructing datasets based on the numerical model simulations for a broader view of network states (Li et al., 2023). However, investigations utilizing high-precision 1/2D bidimensional coupled mechanistic models have predominantly concentrated on the prediction of 2D surface water depths (Chang et al.,

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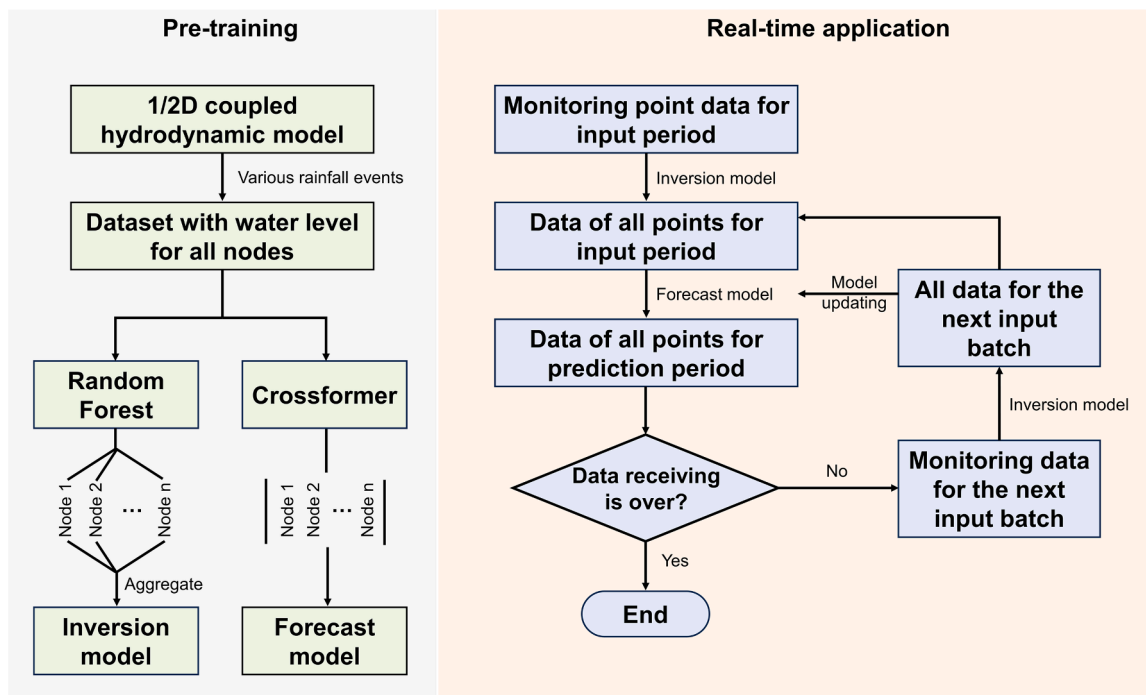


Fig. 1. The framework of the real-time drainage network modeling and updating system.

2022; He et al., 2023; Kim and Han, 2020). Whereas network status studies rely on simpler 1D models (Huang and Lee, 2023; Chiang et al., 2010; Zhang et al., 2024; Zhu et al., 2023), the methodologies employed in these investigations primarily fall into two categories: Recurrent Neural Networks (RNNs) and Graph Neural Networks (GNNs). A shared characteristic among these methods is the requirement for input data related to surface characteristics, such as runoff, or historical global state variables of the network. The acquisition of these inputs poses an additional challenge that merits further study. The data-driven methods aforementioned excel with high-quality data, yet in urban drainage systems, the lack of comprehensive monitoring data presents a significant constraint. While numerical simulations can supplement data availability, the separation of 1D and 2D models for different study objects overlooks their interdependence within the same system, highlighting the need for a unified approach to enhance predictability.

Even with the advancement to higher-quality numerical models, the construction of a sufficiently rich and robust dataset necessitates multiple simulations under various conditions. However, the best a data-driven model trained with simulated data can do is closely mimic the original simulations. Since real-world conditions constantly evolve, simulation data—rooted in static parameters—often fails to align with actual monitoring data, leading to growing discrepancies over time. Consequently, the divergence between predictions and reality may also increase. While data assimilation methods have been explored to reconcile model outputs with real data (Ziliani et al., 2019), they require rigorous observational data, and their complexity, particularly when applied to intricate models, presents a significant challenge for real-time updates. Continuous model updates are essential for effective application, yet research on model updating remains limited. Zhu et al. (2023) explored using transfer learning to update model parameters, but this method implies a direct shift of the learning target to the monitoring data, posing the risk of Catastrophic Forgetting (CF), i.e., losing the previously acquired knowledge (Ao and Fayek, 2023). This approach is particularly precarious when monitoring data is scarce and its reliability cannot be adequately assured.

To tackle the challenges identified, this paper employs a high-precision 1/2D coupled model to construct water level datasets for the entire drainage network, mitigating the inaccuracies inherent in 1D

confluence models when estimating surface runoff inputs at nodes. The resulting high-quality dataset provides a robust foundation for data-driven models. Given the prevalence of water level sensors over flow sensors, pipe flow is not considered in this study. A transformer-based variant of the time series prediction algorithm is used, which demonstrates superior predictive performance through a comprehensive analysis of both temporal and feature scale correlations. The method was rigorously benchmarked against conventional algorithms, while an in-depth analysis was conducted to elucidate the differences in effectiveness between different forecasting patterns. Additionally, the concept of Online Continual Learning (OCL) is introduced as an innovative strategy for real-time model updates, effectively addressing the issue of CF. This research aspires to establish a cohesive and dynamic framework that supports the continuous cycle of model training, implementation, and real-time refinement.

2. Materials and methods

The framework of the proposed real-time drainage network modeling and updating system is illustrated in Fig. 1. In the pre-training phase, a 1/2D coupled hydrodynamic model is employed to simulate various rainfall scenarios, generating a high-precision dataset of nodal water level information. This dataset is then utilized to train both the global state inversion model and the forecast model. The inversion model is trained individually for each target node, with the final results integrated to obtain the global nodal state of the drainage network. In contrast, the prediction model is trained directly for all nodes as a single entity. Upon completion of model training, during the real-time application phase, measured data from monitoring points at the input interval are collected. The inversion model infers the status of all nodes using these data, which are then input as new samples into the forecast model. The forecast model subsequently generates results for the output interval. Concurrently, these new samples also serve to update the forecast model, refining the output for future prediction iterations. This process is repeated throughout the prediction cycle, ensuring continuous improvement in forecasting accuracy.

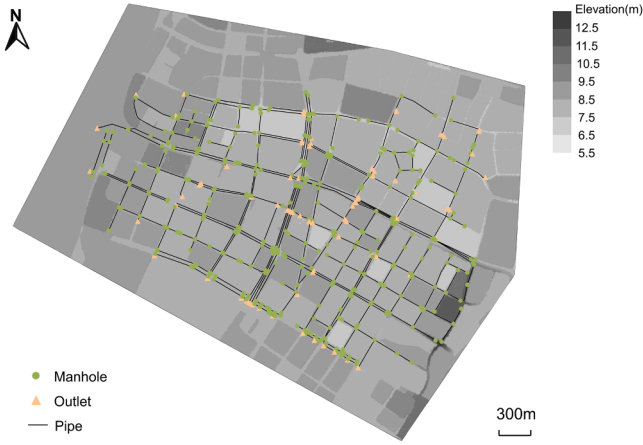


Fig. 2. Topography and drainage network layout of the study area.

2.1. Study area

In this research, we have developed a multivariate time-series prediction model capable of concurrently generating water level data across all nodes, rendering it particularly suitable for deployment in extensive urban drainage systems. A representative real-world urban drainage system situated in the city center of NJ, China, has been selected as the case study. This system spans an area of approximately 18.34 km². Fig. 2 illustrates the system's topography and network configuration.

2.2. Physically-based hydrodynamic model

In this study, a variety of objective rainfall scenarios were meticulously constructed to serve as boundary condition inputs into a robust coupled 1/2D hydrodynamic model, thereby establishing a comprehensive database. The 1D sewer network computational engine harnesses the dynamic wave module of the open-source software SWMM, employing finite difference methods for the numerical solution of the Saint-Venant equations (Rossman and M, 2022). The 2D computational engine is powered by the open-source TELEMAC-2D software, which implements the finite element method to address the SWEs (Hervouet, 2007). The coupling method calculates the interactive flow rates between the two systems at each time step, predicated on the principles of weir and orifice flow equations, and secures bidirectional, tightly coupled interactions via adding point sources or sinks. For further details on the implementation, readers are directed to consult our previous work (Wang et al., 2024). The BC (Bidirectional Coupling) model we developed achieved a satisfactory level of accuracy in a small-scale laboratory test case (Fig. S1), with detailed results provided in the Fig. S2 of supplementary material.

Owing to the paucity of adequate field data for the calibration of model parameters, uniform values were assigned to Manning's roughness coefficients for both land surface and pipes, set at 0.0125 and 0.0128 s·m^{-1/3}, respectively. The drainage network, designed exclusively for stormwater management, comprises 358 manholes, 357 pipes, and 76 outlets. Following a series of simplification procedures, including merging and deletion, the pipes' diameters are large within a range from 500 mm to 2000 mm. The 2D surface modeling employs an unstructured triangular grid with a default resolution of 6 m, encompassing a total of 1,106,759 grids and an average grid area of 16.57m². The rainfall input for the model is assumed to be spatially uniform. The simulation duration is aligned with the temporal span of the rainfall event, with the computational time step set at 5 s and the output time step at 30 s.

2.3. Forecast model

Crossformer (a variant of the Transformer algorithm) is applied for

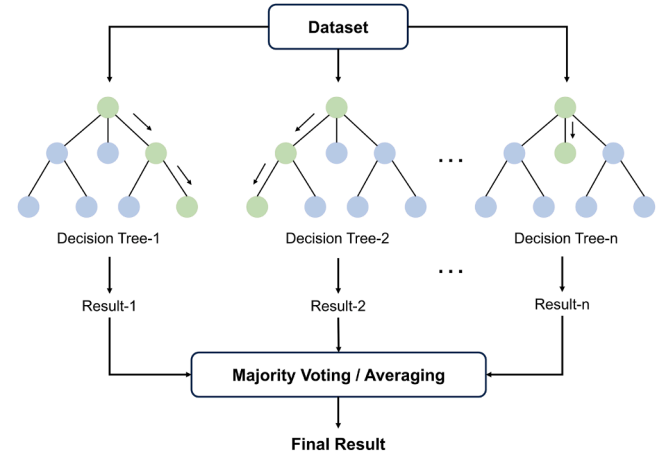


Fig. 3. The structure of the Random Forest algorithm.

the time-series forecasting of nodal water levels within networks in this research. The impact of varying the forecast horizon through different forecast patterns on the accuracy of the prediction outcomes is evaluated.

2.3.1. Inversion of global states from monitoring data

In the present study, the developed predictive model is designed to harness the nodal water levels from preceding time intervals as inputs and to forecast the water levels for future time frames. Nonetheless, the practical deployment of this model encounters limitations due to the finite number of available monitoring points, which cannot encompass the data required for all nodes in the input set. Consequently, this limitation necessitates the implementation of strategies to infer the global states of the network from the restricted monitoring data.

To address the aforementioned limitation, our approach involves the construction of independent regression models for each non-monitored node using the Random Forest algorithm. The Random Forest, an ensemble of decision trees operating under the bagging paradigm, is inherently resistant to overfitting and exhibits robust noise immunity. The architecture of this model is depicted in Fig. 3. The current water level at the monitoring station, the delta from the previous timestep's water level, and the total precipitation are utilized as input variables. The predicted water level at the designated node serves as the output variable. Each time step in the dataset is treated as an independent sample, focusing on establishing data-driven models based on spatial hydraulic characteristic interrelations, as opposed to temporal sequence correlation. The number of nodes excluding the monitoring points corresponds to the quantity of Random Forest models to be developed. The model training for different target nodes can proceed concurrently, with the resultant model parameters being archived in a dedicated repository. So that the pertinent model parameters can be directly invoked during the forecasting task, which can significantly enhance the model's operational efficiency and accuracy.

The layout scheme for the monitoring points was determined through a re-clustering strategy proposed in our previous work (Wang et al., 2023). The potential for re-clustering was indexed by APD as Eq. (1) (the average diameter of the pipes connected to the nodes).

$$APD_k = \frac{\sum_{i=1}^{nk} \sum_{j=1}^{npi} D_i^j}{nk} \quad (1)$$

where APD_k is the APD index of cluster k , nk is the number of nodes belonging to cluster k , npi is the number of pipes connected to node i , D_i^j is the diameter of the pipe j connected to node i .

Starting with an initial cluster count of five, subsequent calculations revealed that these categories did not require further subdivision, as the APD for all five clusters were found to be 1. The final arrangement of the

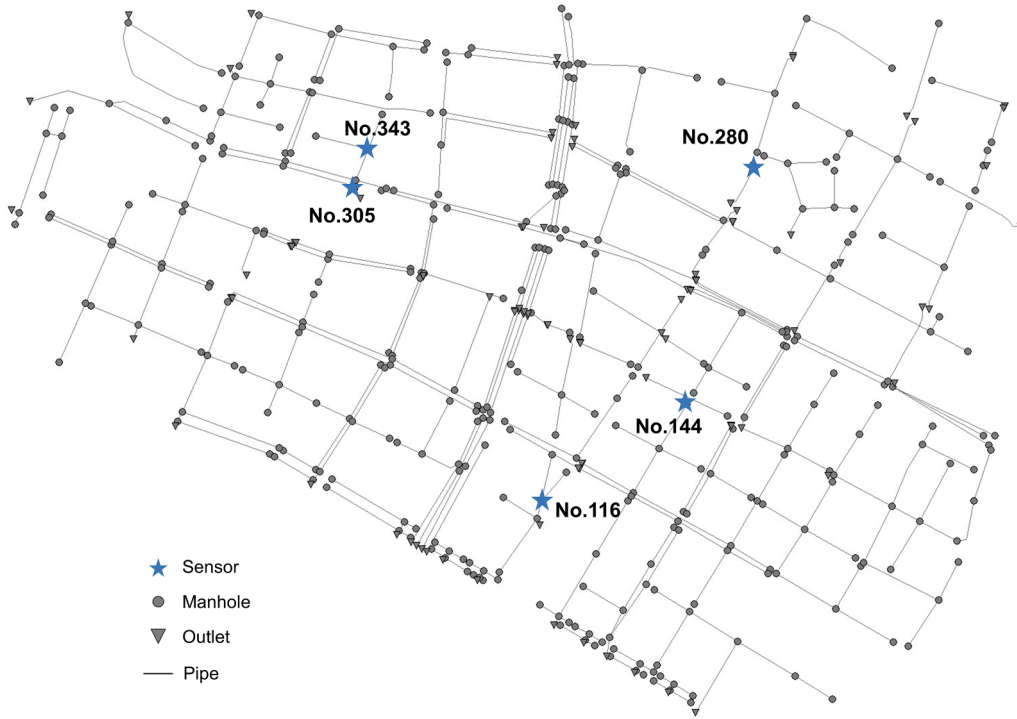


Fig. 4. The layout of monitoring points for the network of the study area.

monitoring points is illustrated in Fig. 4, remains unchanged from the initial clustering. The initial clustering results and the corresponding cluster centers, denoted by asterisks, are presented in Fig. S3 of the supplementary material.

The training dataset comprises 50,760 samples, while the test dataset contains 26,760 samples. The algorithm employs the “Random Forest Regressor” function from the scikit-learn library (Pedregosa et al., 2011) and is implemented in Python 3.9, with parameters set to their default values. Training for the 353 (358-5) target nodes was conducted in parallel across 12 cores.

2.3.2. Crossformer

In recent times, Transformer-based models have emerged as a formidable tool for multivariate time series forecasting, primarily due to their proficiency in capturing long-term dependencies across temporal dimensions (cross-time dependency) (Vaswani et al., 2023). However, these models tend to neglect the interdependencies among various features (cross-dimension dependency), which led to the inception of the Crossformer (Zhang and Yan, 2023). Given the pronounced interconnection between nodal levels within drainage networks, the dataset’s characteristics exactly fit the strength of the Crossformer algorithm. Consequently, it has been selected for the development of the predictive framework in this study.

The Crossformer introduces an innovative Dimension-Segment-Wise embedding approach, where rather than embedding the values of all dimensions of a single timestep into a vector, the points of each dimension are divided into segments of length L_{seg} and then embedded into a vector using linear projection added with a position embedding (Eq. (2)–(3)). It can preserve both temporal and scale information.

$$X_{1:T} = \left\{ X_{i,d}^{(s)} \mid 1 \leq i \leq \frac{T}{L_{seg}}, 1 \leq d \leq D \right\} \quad (2)$$

$$X_{i,d}^{(s)} = \{ X_{t,d} \mid (i-1) \times L_{seg} < t \leq i \times L_{seg} \} \quad (3)$$

where $X_{1:T}$ is the input, $X_{i,d}^{(s)}$ is the i -th segment in dimension d with length L_{seg} , T is the number of time steps in the past, D is the number of

dimensions.

Subsequently, through the Two-Stage Attention layer, it effectively captures dependencies between time steps as well as across different feature scales. First apply multiple self-attention directly to each dimension in Cross-Time Stage:

$$\hat{Z}_{:,d}^{time} = LayerNorm(Z_{:,d} + MSA^{time}(Z_{:,d}, Z_{:,d}, Z_{:,d})) \quad (4)$$

$$Z^{time} = LayerNorm(\hat{Z}^{time} + MLP(\hat{Z}^{time})) \quad (5)$$

where $Z \in \mathbb{R}^{L \times D \times d_{model}}$ is the input of the Two-Stage Attention layer, L and D are the number of segments and dimensions, respectively, d_{model} is the number of dimensions after embedding. $Z_{:,d}$ denotes the vectors of all time steps in dimension d .

The dependency among time segments in the same dimension is captured in Z^{time} , which is the input of Cross-Dimension Stage to capture cross-dimension dependency:

$$B_{i,:} = MSA_1^{dim}(R_{i,:}, Z_{i,:}^{time}, Z_{i,:}^{time}), 1 \leq i \leq L \quad (6)$$

$$\bar{Z}_{i,:}^{dim} = MSA_2^{dim}(Z_{i,:}^{time}, B_{i,:}, B_{i,:}), 1 \leq i \leq L \quad (7)$$

$$\hat{Z}^{dim} = LayerNorm(Z^{time} + \bar{Z}^{dim}) \quad (8)$$

$$Z^{dim} = LayerNorm(\hat{Z}^{dim} + MLP(\hat{Z}^{dim})) \quad (9)$$

where $R \in \mathbb{R}^{L \times c \times d_{model}}$ (c is a small fixed number) is the learnable vector array serving as routers, $B \in \mathbb{R}^{L \times c \times d_{model}}$ is the aggregated messages from all dimensions. $Z_{i,:}$ denotes the vectors of all dimensions at time step i . $\bar{Z}^{dim}$ denotes output of the router mechanism, $\hat{Z}^{dim}$ and Z^{dim} denote the output of skip connection and MLP respectively.

Eventually, the proposed Dimension-Segment-Wise embedding, Two-Stage Attention layer and segment merging are applied to construct a Hierarchical Encoder-Decoder to capture information at different temporal scales.

Table 1
Hyper-parameters configuration in Crossformer.

Hyper-parameters	Values
Epochs	100
Number of encoder layers	2
Number of decoder layers	3
Batch size	32
Dimension of hidden states	500
Number of heads	8
Number of routers of TSA	50
Segment length	5
Learning rate	0.0002
Dropout	0.1

The computational hardware configuration for the experimental platform in this study consists of an AMD EPYC ROME 77,002 CPU and an NVIDIA GeForce RTX 3090 GPU. The Crossformer model is implemented using the Pytorch deep learning library within Python 3.9 (Paszke et al., 2019).

The dataset comprises discontinuous time series formed by concatenating the results of different rainfall events. The time labels are counted in relative seconds rather than in the traditional timestamp format. Compared to conventional long-term time series, the dataset in this study qualifies as a short-term time series, indicating shorter time spans between data points and a more concentrated temporal dimension of information. In light of these characteristics, this research did not employ complex temporal embedding structures; instead, it utilized 1D convolutional layers to construct feature data embeddings, combined with a positional embedding mechanism to aid the model in understanding the sequential relationships among various time points in the series. Given that the shortest rainfall events last only 60 min, the input and output periods set in this study are also relatively brief. The infrastructure is to input the first 20 min (40 timesteps) of inverted node levels and cumulative rainfall data, totaling 359 features, to predict the feature values for the following 10 min (20 timesteps).

Other hyperparameters involved were primarily optimized using the Hyperopt library's Tree-structured Parzen Estimator (TPE) algorithm within a maximum of 50 iterations (Bergstra et al., 2015), with the final hyperparameters presented in Table 1.

2.3.3. Forecast patterns

In the realm of multistep time series forecasting, the performance of predictions varies with the length of the forecast horizon. An extensive literature review has substantiated that, generally, the longer the forecast horizon, the lower the prediction accuracy tends to be (Yang et al., 2024). This study explores the effects of extending the forecast output duration by employing different forecast patterns, aiming to enhance the practicality of the predictive models from perspectives beyond merely altering the forecasting algorithm structure. A schematic of the different forecast patterns is shown in Fig. 5.

Fig. 5(a) illustrates the fundamental structure of multi-step forecasting, where the blue bars represent the input features from previous periods, and the orange bars depict the output features for the predicted temporal segments. Upon accruing a sufficient quantum of new data from the input intervals, the model propels the forecast forward incrementally. Fig. 5(b) extends the forecast horizon at each step by directly increasing the algorithm's output length. Fig. 5(c) extends the forecast horizon via a rolling forecast mechanism, that is, utilizing the output of the antecedent forecast as the subsequent input. Fig. 5(d) delineates the strategy of training an ancillary model for further periods (indicated by the green bars), where the output of the two disparate models is combined at each forecasting step to broaden the temporal scope of predictability. Elaboration on the precise configurations of these patterns will be detailed in the case study section.

2.4. Real-time model updating

2.4.1. Online continual learning (OCL)

The processing of continual learning for both classification and time series regression problems can be expressed in the following mathe-

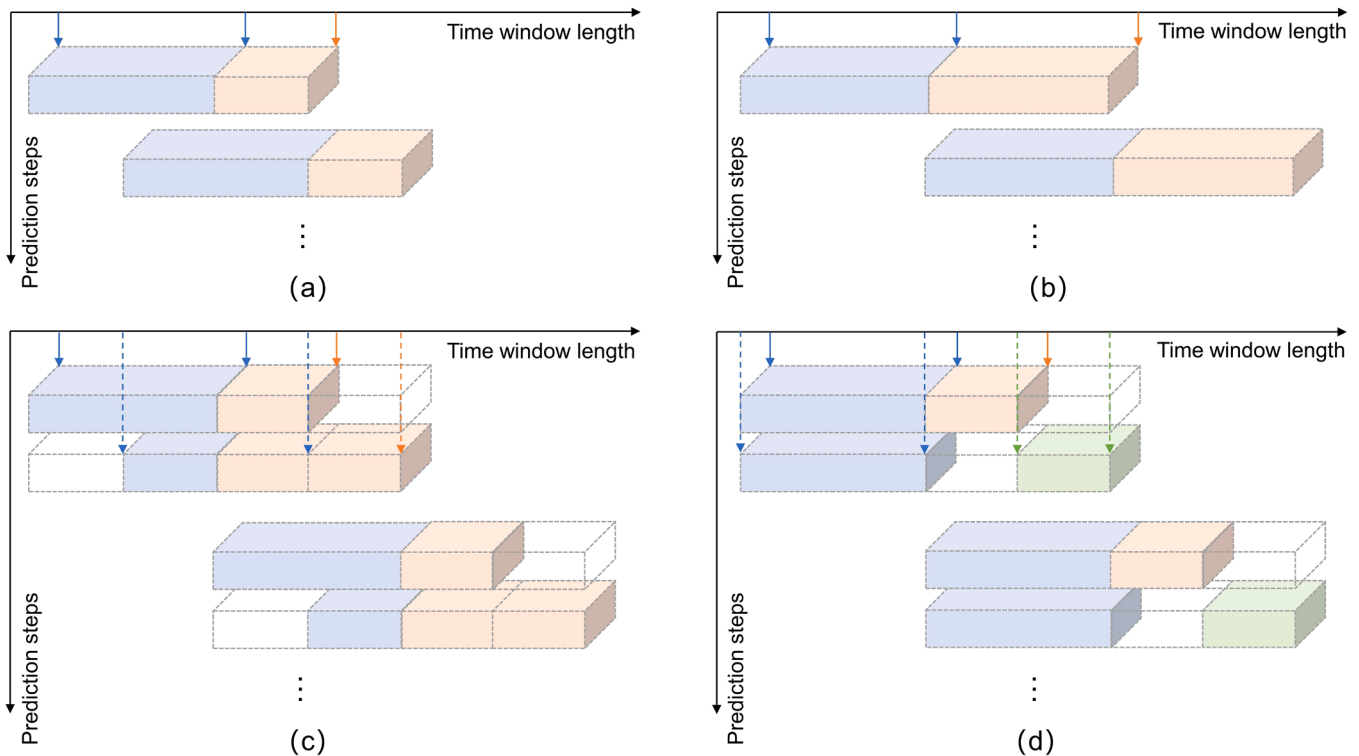


Fig. 5. The schematic of three different forecast patterns: (a) the basic input-output structure; (b) directly increase the output window length; (c) scroll forward forecast; (d) merge different models.

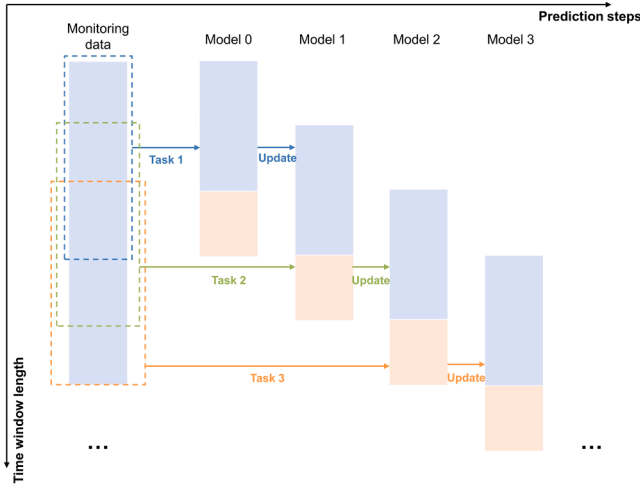


Fig. 6. Process of real-time prediction and model updating.

mathematical form (Gupta et al., 2021): Let $T = (T_1, T_2, \dots, T_m)$ denotes the m tasks that arrive in sequence. There are N instances of labeled time series data $D_i = \left\{ (x_{i,r}, y_{i,r}) \right\}_{r=1}^N$ in the i -th task T_i , where $x_{i,r} \in X_i$ denotes the input and $y_{i,r} \in Y_i$ the corresponding target. Continual learning constrains each task T_i from accessing data of previous tasks $T_j (j = 1, \dots, i - 1)$. The overarching objective for each task is the derivation of a solver model M_i , such that $M_i: X_i \rightarrow Y_i$, parameterized by θ_i and predicts the target as $\hat{y}_{i,r} = M_i(x_{i,r}; \theta_i)$. For time series regression tasks, the squared error loss L serves as the training objective, formalized in Eq. (10) as follows:

$$L_i = \frac{q}{N} \sum_{r=1}^N L(y_{i,r}, \hat{y}_{i,r}; \theta_i) + \left(\frac{1-q}{(i-1)N} \right) \sum_{j=1}^{i-1} \sum_{r=1}^N L(y_{j,r}, M_i(\hat{x}_{j,r}; \theta_i); \theta_i) \quad (10)$$

where $q \in (0, 1]$ quantifies the relative importance assigned to the data from the current task T_i . The term $L(y_{i,r}, \hat{y}_{i,r}; \theta_i)$ and $L(y_{j,r}, M_i(\hat{x}_{j,r}; \theta_i); \theta_i)$ represents the losses incurred on the new and previous tasks, respectively.

OCL is a more challenging and realistic variant of continual learning, wherein training samples for each task arrive from the data stream one by one, with less computational and memory budget. The propensity of OCL to facilitate frequent and efficient model updates aligns with the practical exigencies of real-time model refinement, as necessitated by this study. Consequently, the process of real-time prediction and model updating based on OCL proposed in this paper can be expressed in Fig. 6. The monitoring data are acquired in real-time and inverted to obtain the global state of the drainage network. These data serve as the online training data stream for OCL. Model 0 symbolizes the pre-trained model, as described in Section 2.3. During the online training phase, the model periodically receives new training samples that conform to the length format defined by the pre-trained model. Each new sample is treated as a new task. Each task embodies a fresh set of inputs and outputs, prompting the model to adjust its parameters accordingly (as indicated by the "Update" arrows in Fig. 6). The iteratively updated models (e.g., Model 1, Model 2, Model 3) are then used for subsequent prediction tasks. Collectively, this framework guarantees the utilization of the most up-to-date model parameters for the prediction of the network's future states, thereby augmenting the model's precision and adaptability.

The data stream within the OCL framework consists of nodal water levels and cumulative rainfall data. Since the monitoring equipment has not been deployed in the study area, this study compensates for the absence of actual measured nodal water level data at the monitoring

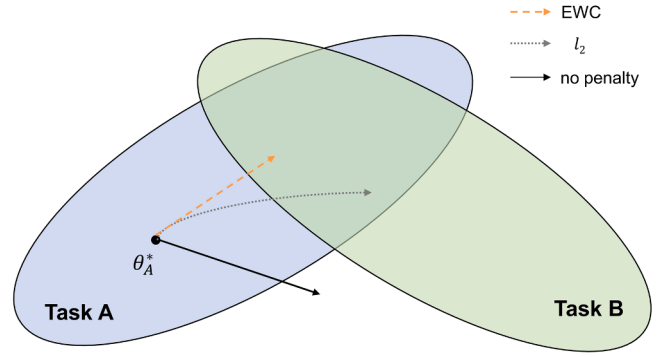


Fig. 7. Illustration of training trajectories in EWC.

sites. This compensation involves the generation of new rainfall events not present in the training set, the adjustment of numerical model parameters to simulate these events, and the addition of a predetermined level of noise to the simulated results, thereby substituting for the unobtainable actual monitoring data. The sample format is formatted to align with the predictive model's requirements. The training data stream generates a sample every 10 min as a new task. During the training process of each new task, "fast training" is first conducted on the current sample using a learning rate reduced to 10 % of the initial rate. This step is designed to quickly acclimate the model to the latest data. Then, the sample is incorporated into the memory pool, where it undergoes extensive training across multiple cycles with the existing samples. To prevent unbounded growth in the model's memory footprint during operation, the memory pool is capped at a predetermined maximum (300 samples in this instance). Samples exceeding this cap are randomly pruned to maintain the limit. Considering that the learning rate of the pre-trained model is already markedly low when it converges, the initial learning rate for OCL is set to 1×10^{-7} .

2.4.2. Elastic weight consolidation (EWC)

In the continual learning framework, academics have proposed a spectrum of polynomial time heuristic algorithms to overcome the CF problems. These algorithms can be classified into three categories in terms of their approach toward the storage and utilization of task-specific information throughout sequential learning processes, i.e., regularization-based methods, memory replay methods, and parameter isolation methods (Delange et al., 2021). Among these, the regularization method achieves to acquisition of the new task while maintaining the existing knowledge by imposing constraints on weight adjustments during model updates. Its lower memory requirement and superior computational efficiency render it particularly advantageous for extended online learning scenarios. Within this study, the Elastic Weight Consolidation (EWC) algorithm, a prime exemplar of regularization-based algorithms, has been selected for detailed examination.

The primary objective of EWC is the retention of proficiency in task A while training on task B, as shown in Fig. 7. This figure illustrates the training trajectories within a schematic representation of the parameter space, highlighting parameter zones that yield optimal performance for task A (blue) and task B (green). Following the acquisition of task A, the model parameters are positioned at θ_A^* . If gradient steps (gray dotted arrows) are taken based solely on task B's data, it might minimize task B's loss at the expense of eradicating the knowledge previously acquired for task A. Conversely, applying uniform constraints across all weights (black solid arrows) results in overly stringent limitations, hindering the model's ability to learn task B efficiently. EWC navigates this dilemma by valuing the significance of a trade-off parameter for task A, thereby enabling the acquisition of task B's knowledge without substantially impairing performance on task A (orange dashed arrows).

The loss function needed to be minimized in EWC is rearranged as

Eq. (11):

$$L(\theta) = L_B(\theta) + \sum_i \frac{\lambda}{2} F_i (\theta_i - \theta_{A,i}^*)^2 \quad (11)$$

where $L_B(\theta)$ is the loss for task B only, λ is the weighting adjustment factor, i labels each parameter, F_i represents the parameter importance on task A (old task), which means the diagonal of the Fisher information matrix (Kirkpatrick et al., 2017). It can be calculated by Eq. (12):

$$F_i = \frac{1}{|D|} \sum_{d \in D} \frac{\partial^2 L(d, \theta)}{\partial \theta^2} \quad (12)$$

where D is the dataset, $L(d, \theta)$ is the loss of a data sample d for the model with parameter θ .

Through testing, it has been observed that the regularization loss term's weight parameter exerts a small effect on the final updated model's performance, and is therefore consistently configured to a value of 1 in this case.

2.5. Rainfall events

To enhance the model's generalization capabilities, it is essential to simulate operating scenarios across a variety of weather conditions. For short-duration rainfall events, the construction of single-peak rainfall profiles employs the Chicago rainfall pattern (Arnell et al., 1984), and the rainfall intensity equation after applying local parameters for NJ city is shown in Eq. (13). The formulation of the rainfall process line incorporates three critical parameters: the rainfall return period (P), the peak rainfall coefficient (r), and the rainfall duration (T). For this study, the design rainfall return period is set between 2 and 100 years, the duration ranges from 60 to 180 min, and the peak rainfall coefficient varies from 0.2 to 0.8. To more accurately mirror real-world conditions and increase the sample size, 10 % to 50 % Gaussian white noise was added to the design rainfall data, resulting in a comprehensive dataset of 450 single-peak rainfall scenarios.

$$q = \frac{10716.7(1 + 0.837 \lg P)}{(t + 32.9)^{1.011}} \quad (13)$$

where q is the rainstorm intensity ($L \cdot s^{-1} \cdot hm^{-2}$), P is the rainfall return period, t is the rainfall duration (min).

In addition, bimodal rainfall patterns of extended duration and certain historical instances of exceptionally heavy rainfall were also considered. These bimodal patterns were established utilizing local historical rainfall data, in alignment with the methodology described by Pilgrim & Cordery method (Pilgrim and Cordery, 1975). The considered rainfall durations for these events ranged from 240 to 720 min, with return periods spanning 2 to 100 years. Additionally, the analysis also encompassed extreme historical rainfall events of even longer durations, specifically up to 1080 and 1440 min, to capture a comprehensive range of scenarios. A total of 504 distinct rainfall events were constructed to form a robust dataset. This dataset was systematically partitioned into training, validation, and test subsets following a 7:1:2 proportion, with the test subset containing 103 distinct rainfall events.

Table 2

Inversion model representation of target nodes.

Statistical values	MAE(m)	RMSE(m)
Average	0.003	0.014
Median	0.003	0.004
Max	0.072	0.273
Min	0.000	0.000

2.6. Evaluation

In this study, the evaluation metrics employed include Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Nash-Sutcliffe Efficiency (NSE), and the correlation coefficient (CORR), with their mathematical formulations presented as Eq. (14)–(17):

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_{pred,i} - y_{true,i}| \quad (14)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{pred,i} - y_{true,i})^2} \quad (15)$$

$$NSE = 1 - \frac{\sum_{i=1}^N (y_{pred,i} - y_{true,i})^2}{\sum_{i=1}^N (y_{pred,i} - \bar{y}_{pred})^2} \quad (16)$$

$$CORR = \frac{\sum_{i=1}^N (y_{pred,i} - \bar{y}_{pred})(y_{true,i} - \bar{y}_{true})}{\sqrt{\sum_{i=1}^N (y_{pred,i} - \bar{y}_{pred})^2 \sum_{i=1}^N (y_{true,i} - \bar{y}_{true})^2}} \quad (17)$$

where N is the number of samples, $y_{pred,i}$ is the model output value, $y_{true,i}$ is the target output value, $\bar{y}_{pred}$ is the mean of model outputs and $\bar{y}_{true}$ is the mean of target outputs.

For the assessment of the random forest model, metrics that represent absolute numerical differences, such as MAE and RMSE, are primarily utilized. In contrast, the evaluation of the predictive model and model updating strategy necessitates the use of NSE and CORR, which reflect the degree of alignment with temporal trends.

The predictive model is assessed from two perspectives regarding the test set outcomes: Firstly, for each rainfall event, the predicted time series for each node under that particular event is compared with the actual time series, from which the evaluation metrics are calculated. The average of these metrics is used to represent the overall prediction error for the rainfall event. It is noteworthy that due to the minimal water level changes at some nodes, NSE may exhibit exceptionally low outlier values. To mitigate the impact of such cases on the overall error assessment, nodes with a range of change < 0.05 m throughout the rainfall event are disregarded when calculating NSE. Secondly, for each node, its overall performance across all rainfall events in the test set is calculated. Specifically, the concatenated predicted time series for each node across all events is compared with the actual time series to compute the evaluation metrics. Additionally, this study engaged in a comparative analysis of the Crossformer algorithm against a cadre of conventional time series forecasting algorithms, including LSTM (Hochreiter and Schmidhuber, 1997), the foundational Transformer algorithm, and its variants, PatchTST (Nie et al., 2023) and iTransformer (Liu et al., 2023).

The assessment of the model updating strategy used a long-duration rainfall event with actual monitoring data as a preparatory data stream. The global state inferred from monitoring data serves as the target value. The error in the predicted time series under the OCL strategy is calculated to evaluate the performance of new tasks. Moreover, the updated model is employed on the old test set of the pre-trained model to assess its ability to resist the CF problem. Finally, the practicality and reliability of the model updating strategy applied in this research are verified by comparing it with two alternative approaches: one without model updates and the other using a fine-tuning strategy of transfer learning (only the last layer of decoder parameters are retrained).

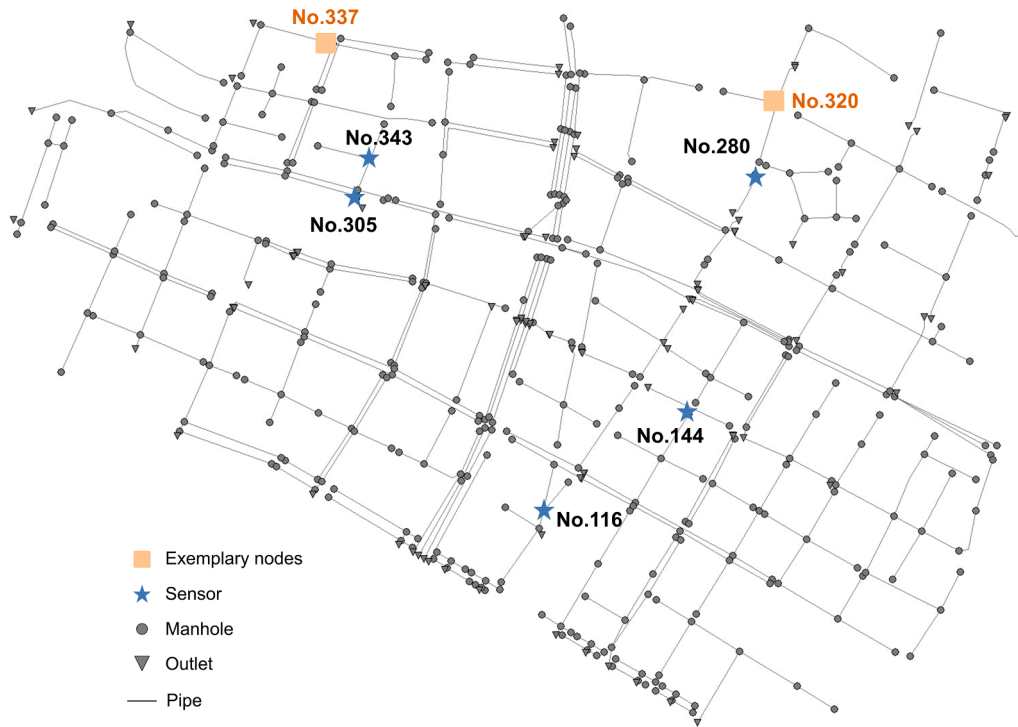


Fig. 8. Locations of the exemplary nodes for the inversion model.

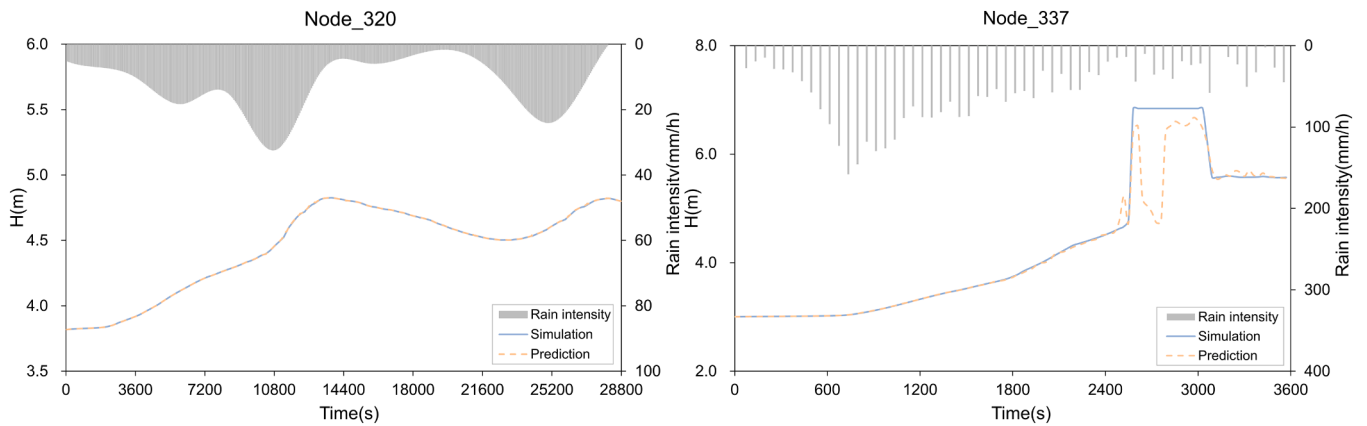


Fig. 9. Inversion results for the representative nodes during different rainfall events.

3. Results and discussion

3.1. Global state inversion

Table 2 presents the outcomes obtained from evaluating the trained random forest models across 353 designated target nodes. Comprehensive results for each individual node are provided in Table S4 of the supplementary material. Notably, the average values of the MAE and the RMSE are observed to be consistently low, indicative of the models' robustness and precision. To further validate the inversion results, we sampled two types of typical nodes with different performance from disparate regions of the drainage network (Fig. 8 delineates their spatial distribution) and examined their performance under various testing rainfall events (Fig. 9). Node 337, characterized by the highest MAE and RMSE values, was among the selected nodes. It is evident that the majority of nodes, such as the Node 320 presented, achieved a high level of accuracy in the inversion results. However, nodes with larger errors, similar to node 337, experienced extended periods of overflow

conditions during simulation. The minimal fluctuations in water level during these overflow states present a formidable challenge for the Random Forest model to enact precise decision regression, leading to heightened error margins in these cases. Nevertheless, the model's consistency in capturing the pivotal moments and overall trends of change fulfills the fundamental objectives of the inversion process. This finding substantiates the capability of the inversion model to precisely deduce the water level states at points lacking direct monitoring, thereby indirectly affirming the strategic selection and high representativeness of the monitoring locations employed in this study.

3.2. Forecast model

3.2.1. Base structure performance

In the subsequent section, we delve into the performance of the forecast model premised on a fundamental input-output structure. Besides MAE and RMSE, NSE and CORR metrics are also introduced into the evaluation system. The objective is to ascertain the applicability of

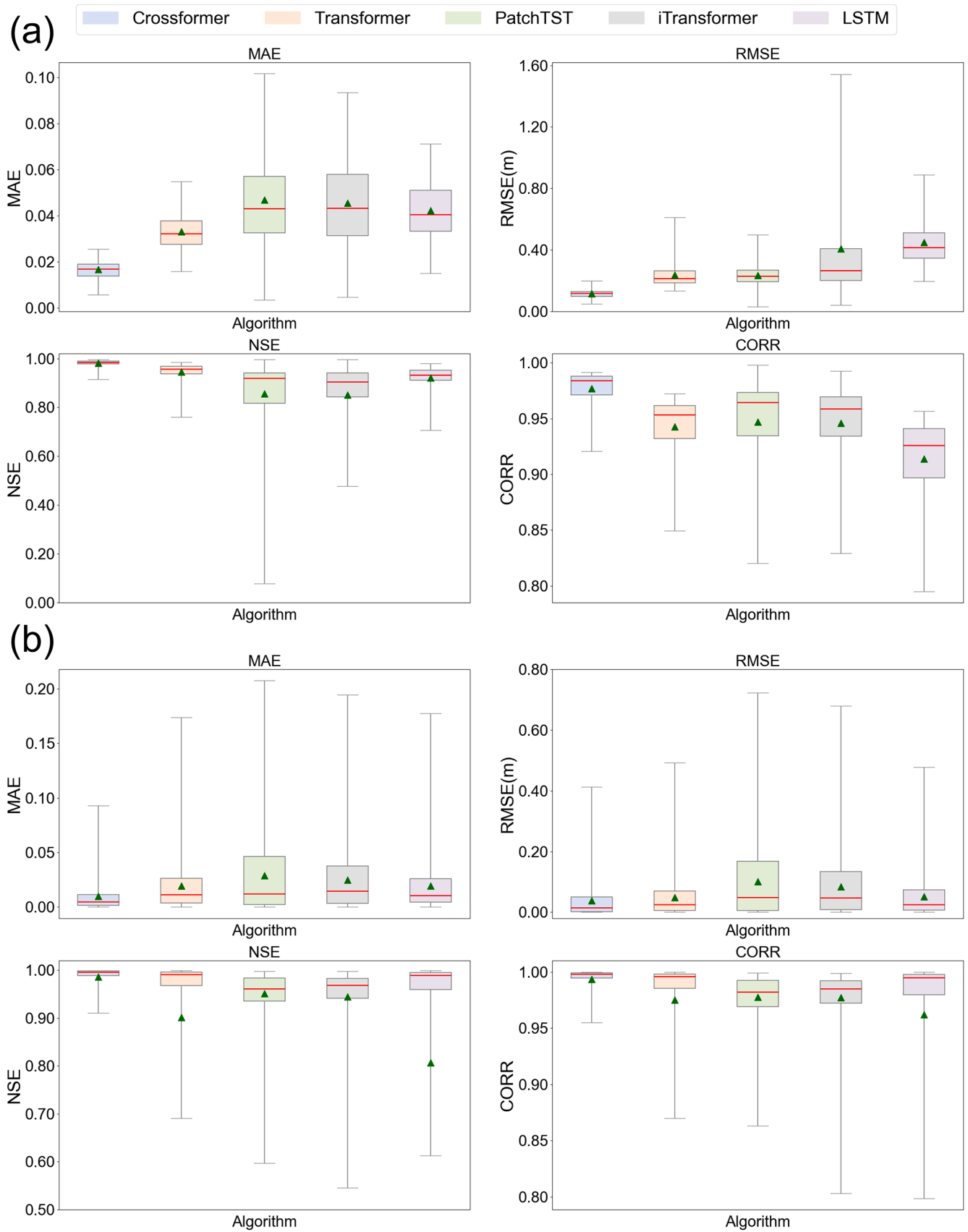


Fig. 10. Boxplot of prediction error distribution with different algorithms: (a) across different rainfall events; (b) across different nodes.

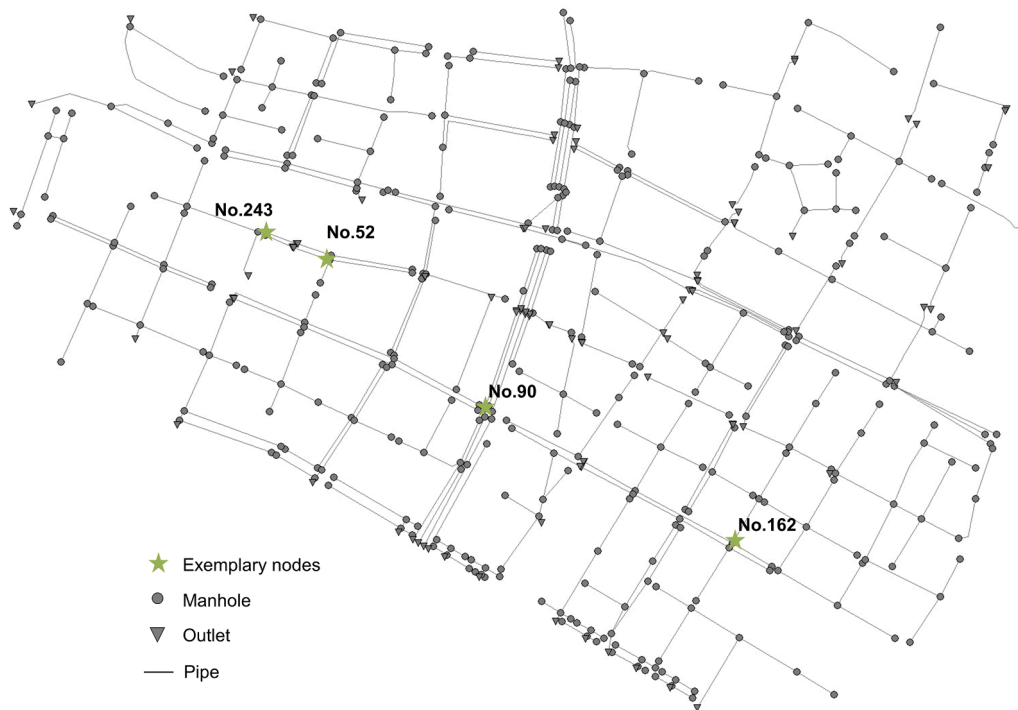


Fig. 11. Locations of the exemplary nodes for the forecast model.

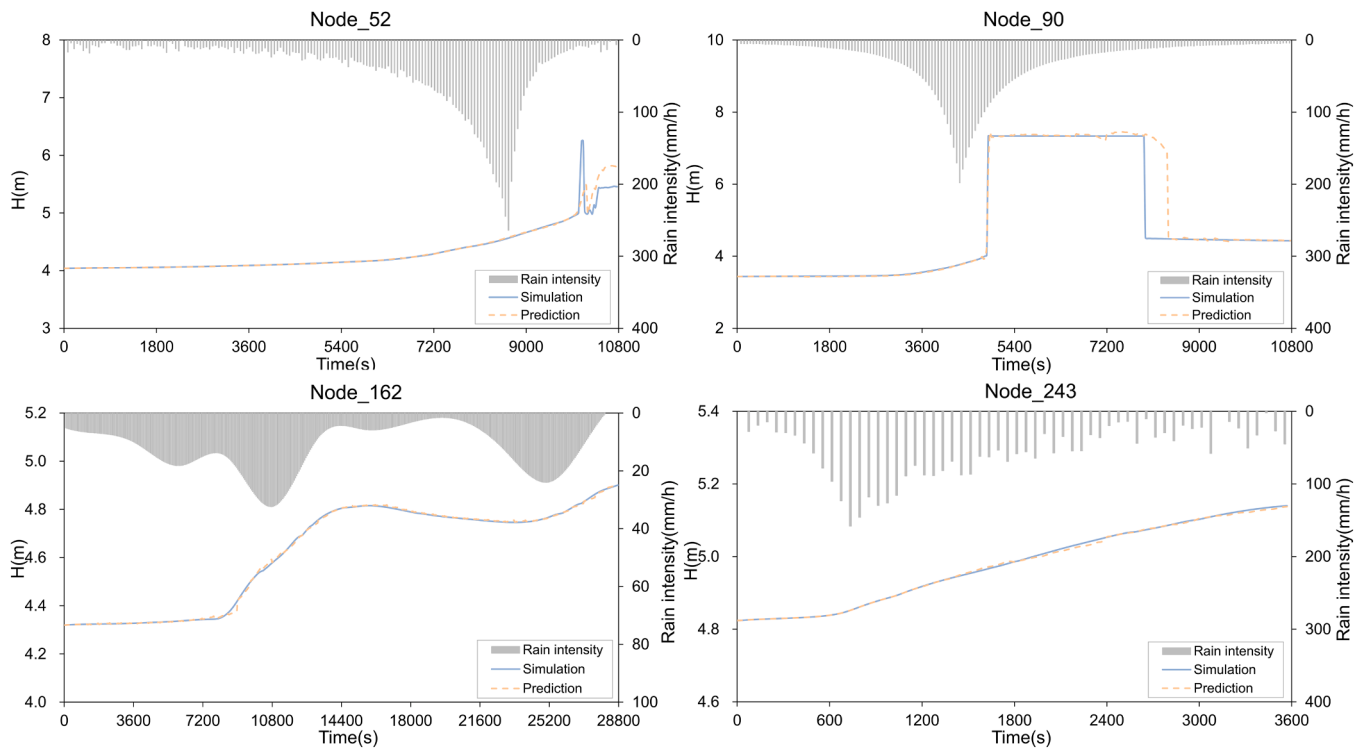


Fig. 12. Prediction results for the representative nodes during different rainfall events.

the Crossformer algorithm within the context of this case study.

For the test set encompassing various rainfall events, Fig. 10(a) presents a boxplot illustration of the error distribution for each rainfall event (whiskers extend to 10 times the interquartile range, excluding data beyond as outliers, same applies subsequently; the horizontal line represents the median and the triangles the mean). Fig. 10(a) demonstrates that the Crossformer algorithm achieves the most favorable

overall prediction outcomes across 103 test set rainfall events, surpassing the other algorithms across all evaluated metrics. Furthermore, Crossformer exhibits a notable advantage in terms of error dispersion, suggesting its ability to maintain prediction precision across different scenarios. The prediction efficacy across different rainfall events is generally commendable, with the disparities in predictions for individual events being minimal and not showing a clear pattern, i.e., there

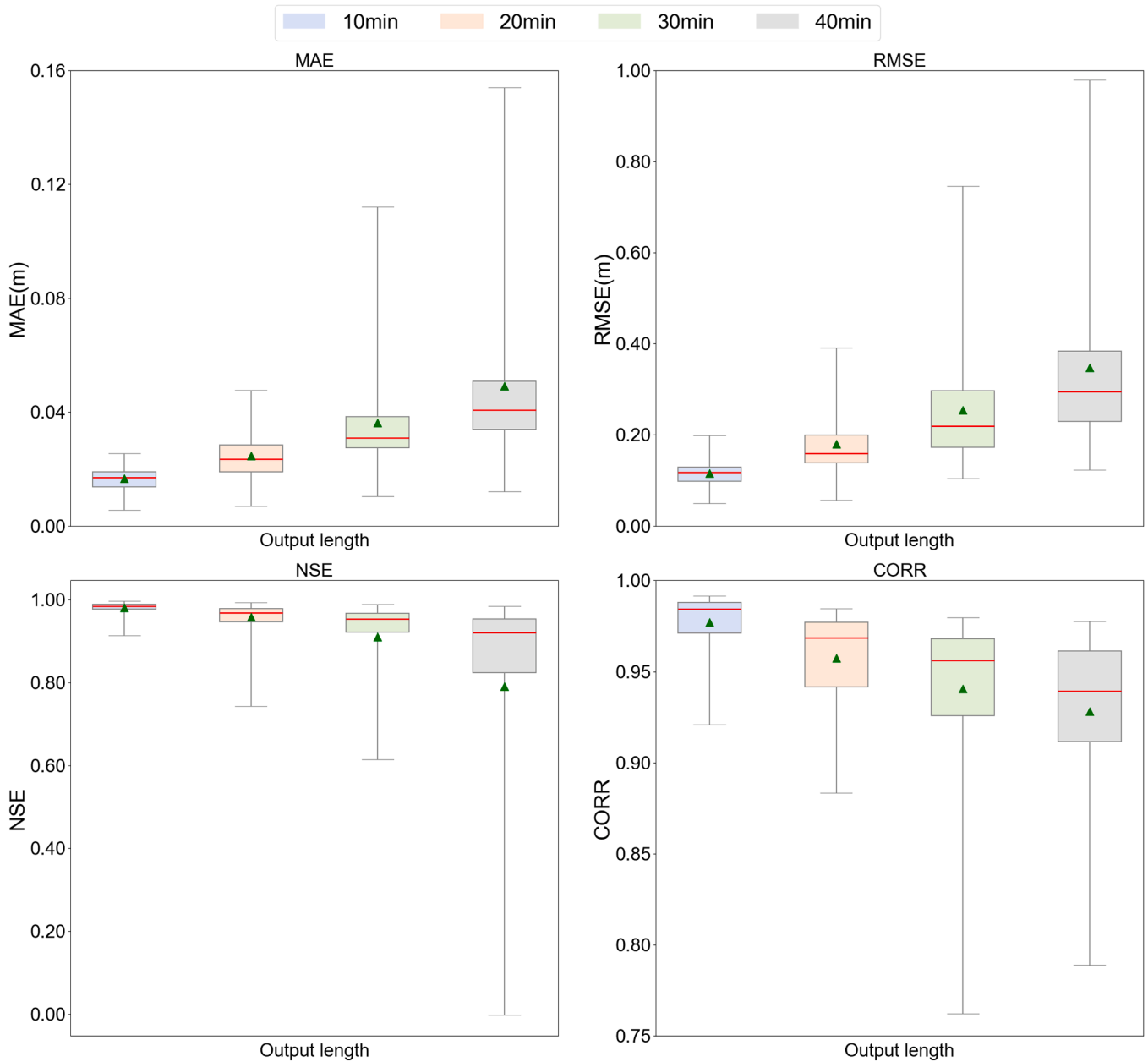


Fig. 13. Box plots of prediction error distribution with different forecast time lengths.

appears to be a stochastic correlation between these differences and factors such as rainfall duration, recurrence intervals, and noise percentages.

Fig. 10(b) presents a boxplot of the error distribution for the collective performance of all nodes across the full set of test rainfall events. It is clear from Fig. 10(b) that the Crossformer algorithm consistently delivers superior predictive performance across all nodes, with minimal dispersion in errors. In contrast, the other algorithms exhibit localized deficiencies in certain nodes. Despite the predominantly superior predictive performance across the majority of nodes, we observe a certain degree of performance disparity among different nodes. To facilitate a more in-depth analysis, this study selects four representative nodes (their locations were marked in Fig. 11) that exhibit a range of performance levels to closely examine the water level prediction results for specific test set rainfall events (Fig. 12).

As shown in Fig. 12, Nodes 162 and 243 are selected as exemplary in terms of satisfying prediction performance, accurately capturing the temporal trends of both short-duration and long-duration rainfall

events. This level of accuracy is consistent across over 90 % of the nodes, indicating the model's high predictive capability on these nodes. Node 52 is selected as an example with a slightly lower NSE index (0.918). The suboptimal NSE is primarily due to fluctuations and abrupt changes associated with the numerical computation processes of hydrodynamic model simulations, making it difficult for the model to predict such irregular changes in detail. However, the overall trend is still well-captured. Node 90, identified for its slightly poorer RMSE index (0.413), is mainly affected by the lag in the timing of the sudden drop (linked to the handling of transitions between different manhole states by the SWMM's EXTRAN module) in water level. In urban flooding modeling, emphasis is placed on the general trends of manhole water level changes, their magnitude, and overflow risk, rather than the detailed transitions between network states. Therefore, despite encountering localized predictive hurdles, the forecast model developed herein is deemed accurate and reliable, offering substantial reinforcement for the simulation and oversight of urban drainage systems.

It is noteworthy that the performance of other algorithms does not

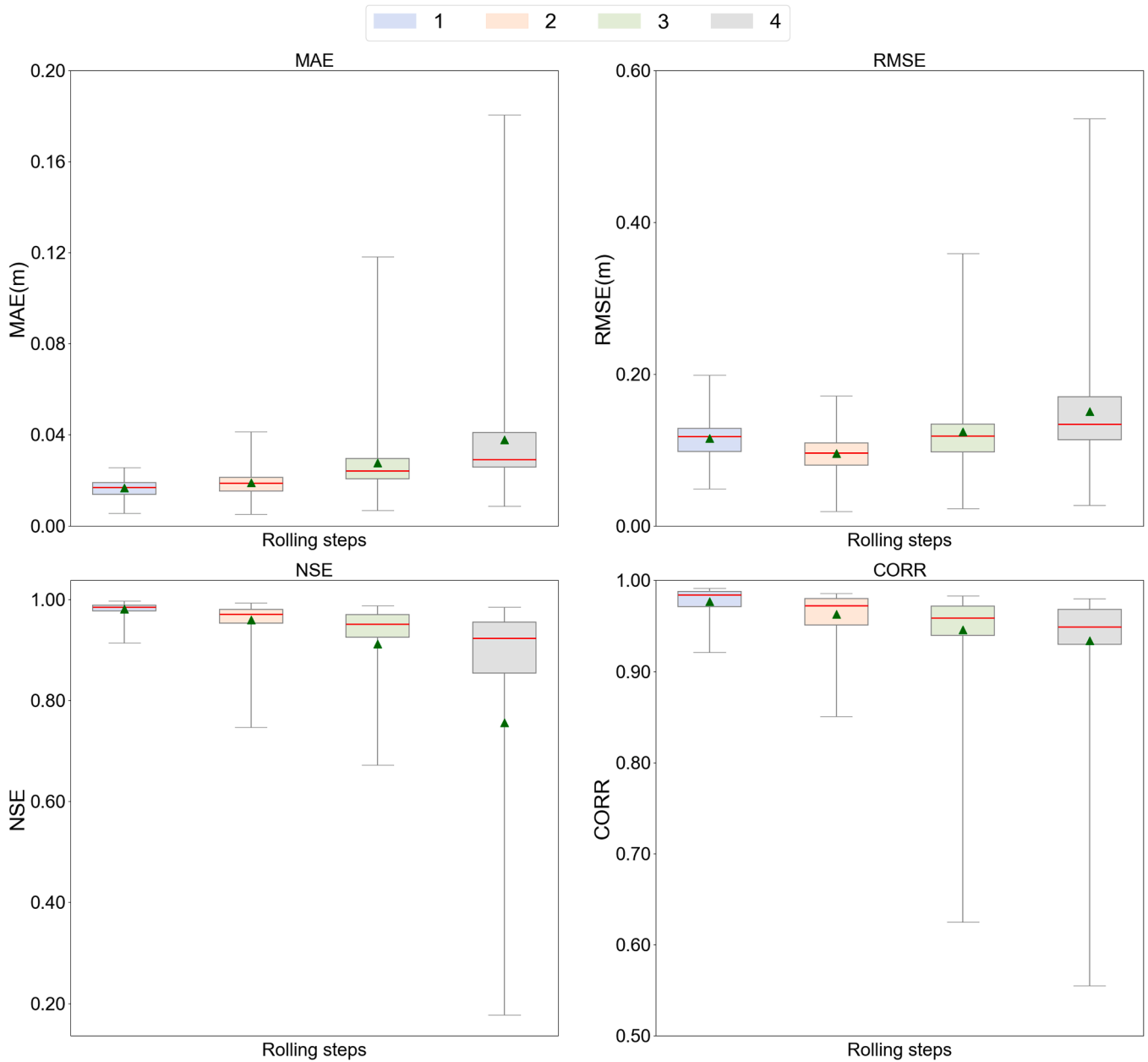


Fig. 14. Box plots of prediction error distribution with different rolling forecast steps.

correlate absolutely with their novelty. For instance, iTransformer and PatchTST, which are more recent variants than Crossformer, have achieved state-of-the-art (SOTA) performance on standardized open datasets. However, in this case study, their performance was even surpassed by the most fundamental Transformer algorithm. This observation is consistent with findings from other studies. For instance, [Vatanchi et al. \(2023\)](#) found that simpler ANN and ANFIS models can outperform more complex models like BiLSTM and CNN-GRU-LSTM in predicting daily runoff. In a similar vein, [Yang et al. \(2024\)](#) conducted a comparison of GRU, LSTM, and BiLSTM algorithms for flood depth prediction revealed that the simpler GRU algorithm was the most effective. These findings underscore that different dataset characteristics necessitate distinct algorithmic structures, and selecting the appropriate tool requires a comprehensive consideration of various factors, including the specific characteristics of the dataset and the computational resources available.

3.2.2. Comparative analysis of different forecast patterns

In this section, we delve into the analysis of outcomes under different

forecasting patterns. At first, following the idea prevalent in most studies, the influence of different forecast durations and varying numbers of rolling forecast steps on predictive performance was examined. The model's performance across various rainfall events within the test dataset is illustrated in [Figs. 13 and 14](#). [Fig. 13](#) distinctly depicts a progressive deterioration in model predictive accuracy as the output length of the forecast model increases. [Fig. 14](#) reveals a trend akin to that observed in [Fig. 13](#); however, it exhibits a lesser degradation in results upon extending the prediction window. This observation implies that rolling forecasts possess enhanced robustness compared to merely elongating the forecast duration, particularly when the quantity of rolling steps is constrained to a small count.

Subsequently, we compared the three different prediction patterns for extending the forecast duration proposed in [Section 4.3.3](#), with the results depicted in [Fig. 15](#). "Direct" refers to directly extending the output window length to 20 min, "Combine" involves amalgamating the results of two models with a 10-minute output each, and "Rolling" refers to a two-step rolling forecast. Each of these methods signifies that each

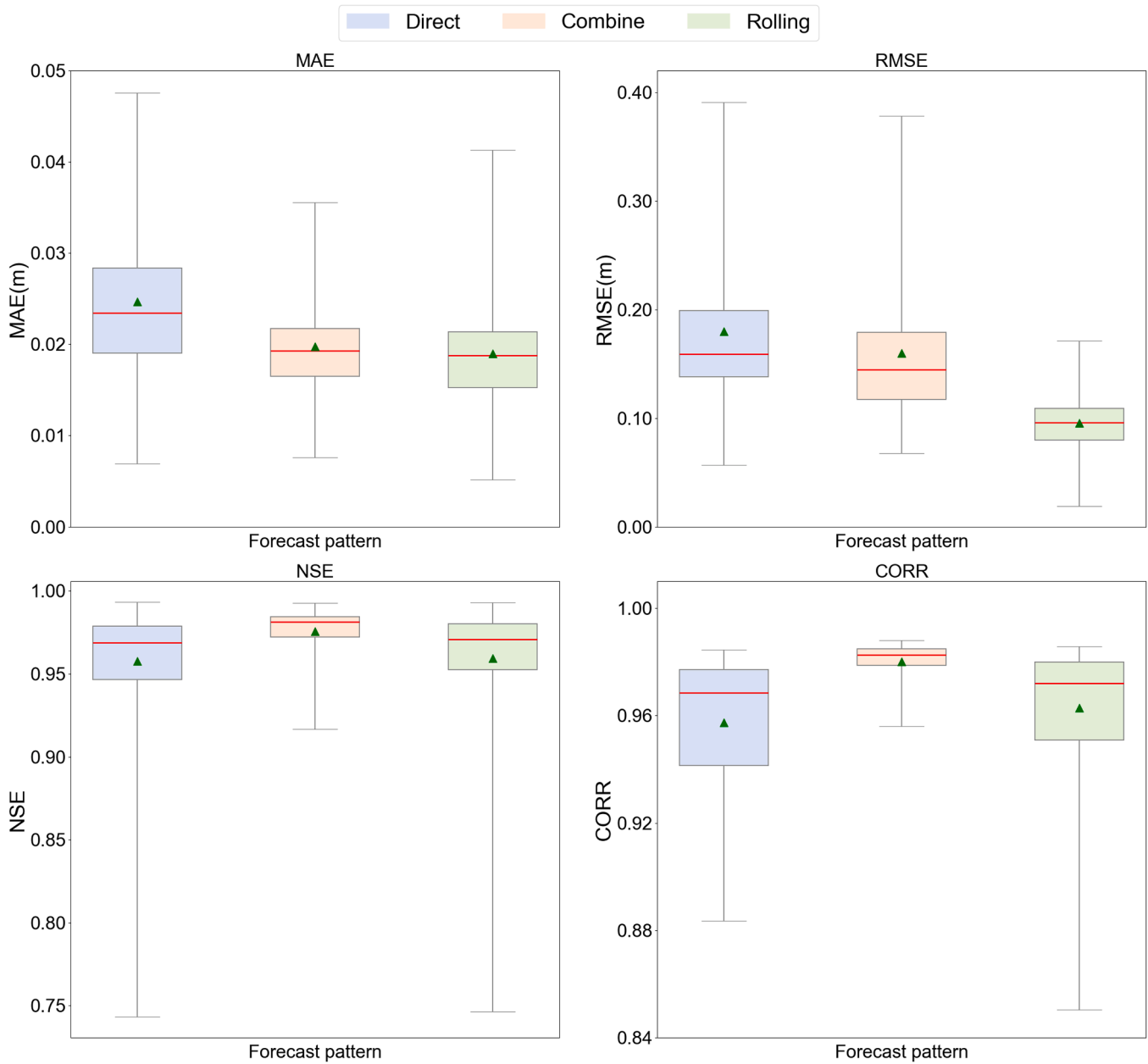


Fig. 15. Box plots of prediction error distribution with different forecast patterns.

Table 3
The predictive accuracy of three different model updating strategies.

Forecast pattern	MAE	RMSE	NSE	CORR
Input 20 min Output 20min	0.025	0.180	0.958	0.957
Input 40 min Output 20min	0.024	0.162	0.868	0.908
Combine	0.020	0.160	0.976	0.980
Rolling	0.019	0.096	0.960	0.963

prediction step outputs 20 min of results. It is apparent that although all three methods yield results with errors within an acceptable range, the rolling forecast method has an advantage over directly increasing the output length in terms of performance in both the MAE and RMSE metrics, which indicate the error of the numerical values. While the method that combines the outputs of two models excels in the NSE and CORR metrics that characterize data trend variation. These findings suggest that directly increasing the output length is not the optimal approach for extending the forecast period. Additionally, we expanded

the experiment by increasing the model structure from an input of 20 min to 40 min, maintaining an output of 20 min; the average results are shown in Table 3. The structure with a 40-minute input shows improvement in the MAE and RMSE metrics compared to the 20-minute input structure, but it performs worse in the NSE and CORR metrics and cannot surpass the other two prediction patterns. This further substantiates that directly altering the algorithm's input-output structure is not as effective as utilizing a well-validated foundational structure.

3.3. Model updating

The results related to the model updating strategy are analyzed and discussed in this section. To incorporate a broader array of new samples and effectively evaluate the impact of distinct model updating approaches, three new rainfall events of considerable duration (240 min, 480 min, and 720 min, respectively) were selected. Utilizing monitoring data from the constructed observatories to inversely deduce the global state, a real-time data stream for each event was generated. Employing a

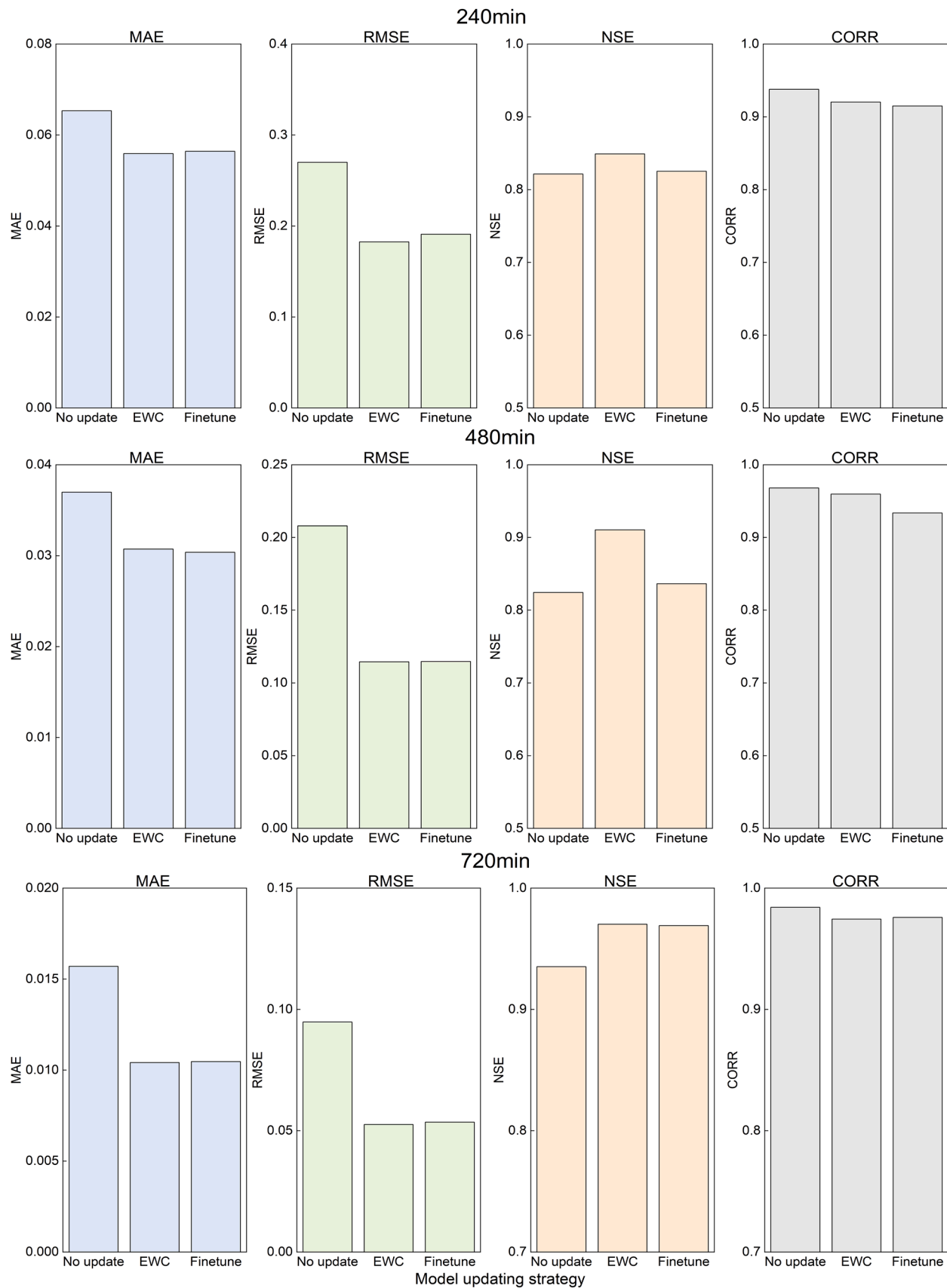


Fig. 16. The performance on new tasks by three different model updating strategies.

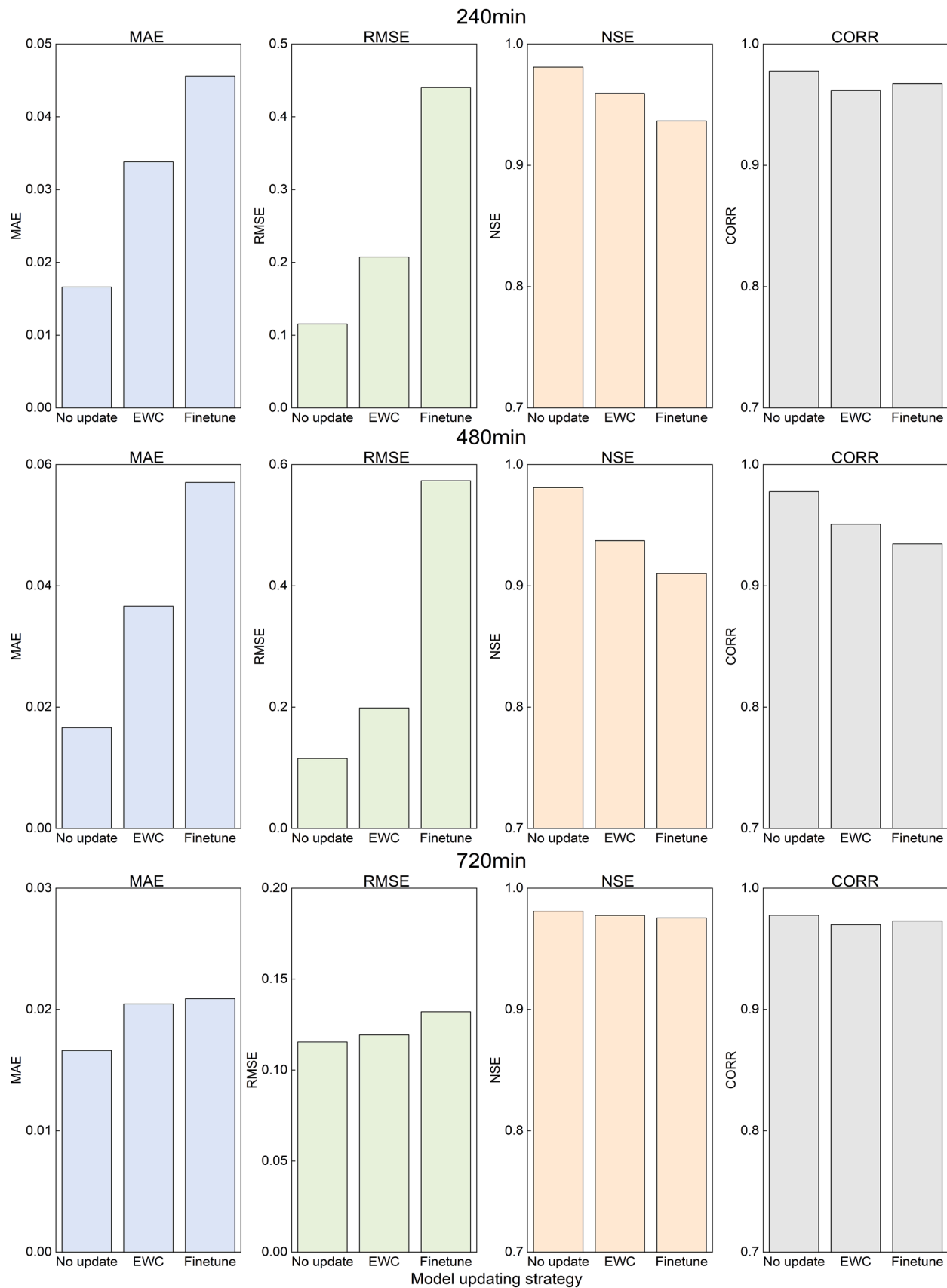


Fig. 17. The performance on old tasks by three different model updating strategies.

forecast model characterized by a basic input-output structure as the pre-trained model, we compared the predictive accuracy of three different strategies: no update, update via fine-tuning, and update using the EWC algorithm proposed in this paper, as illustrated in Fig. 16. A recurring observation across these three rainfall events is the enhancement in predictive performance achieved by both fine-tuning and EWC strategies in comparison to the non-updated model (a slight decrease in CORR may be attributed to the frequent model updates slightly disrupting the data variation continuity), affirming the necessity and effectiveness of model updating. Furthermore, the predictive performance achieved with the EWC strategy outperforms that of fine-tuning, demonstrating the superiority of the model updating strategy adopted in this research for tackling new tasks. A comprehensive examination of different rainfall events reveals that for events where the pre-trained model exhibits suboptimal predictive accuracy, the EWC strategy exhibits a more pronounced advantage. Conversely, for events well-handled by the pre-trained model, the relative advantage is less evident (e.g., the 720-minute event). This observation also provides context for future investigations into assessing the appropriateness of new data for model updates.

Not updating the model circumvents the issue of catastrophic forgetting entirely, thus serving as an optimal baseline for comparison. The efficacy of the EWC and fine-tuning strategies in countering CF issues was evaluated by comparing their performance on an old test set, as outlined in Fig. 17. It is evident that, although both the EWC and fine-tuning strategies inevitably lead to a certain level of old knowledge forgetting, the extent of forgetting with EWC is significantly lower than that with fine-tuning, ensuring that the degradation in performance on the old test set remains within a more controllable limit. Further comparison across different rainfall events reveals that for the 720-minute rainfall event, which has previously been demonstrated in the preceding discussion to involve a lesser degree of updating, both strategies accordingly exhibit a lower degree of forgetting, thereby the disparity between the two strategies is less pronounced. In summary, the model updating framework developed in this study demonstrates exceptional performance in both integrating new knowledge and preserving old knowledge.

4. Conclusions

This research proposes an integrated framework that merges robust monitoring, forecasting, and updating mechanisms, aiming to address the intricate challenges of real-time urban drainage network modeling. The framework receives real-time monitoring data to invert the global state, inputs the prediction model, and predicts the nodal liquid level in the future period. At the same time, the new measured data effectively updates the prediction model to continuously improve the prediction accuracy.

In the construction of prediction models, the utilization of a high-precision 1/2D coupled hydrodynamic model has facilitated the generation of an extensive and accurate training dataset. The integration of the Crossformer algorithm cleverly captures the correlation across temporal and feature dimensions. The findings indicate that this algorithm significantly improves the simultaneous prediction of water levels at various nodes. Specifically, the average MAE is maintained at an exceptionally low level of 0.010 m and the NSE has reached 0.987, outperforming traditional time-series prediction algorithms and other variants of the Transformer algorithm. This study also conducted a thorough comparison of different forecast patterns, clearly demonstrating that augmenting a high-precision basic infrastructure is significantly superior to direct algorithmic restructuring.

Furthermore, this paper incorporates online continual learning into the predictive model for real-time model updating, and the results show that the applied updating strategy performs satisfactorily on both established and emerging scenarios. Specifically, in new task scenarios, the updated model exhibits a nearly 50 % reduction in RMSE compared

to the non-updated model and a significant improvement in NSE over the finetune strategy. Regarding old task scenarios, the RMSE remains below a 0.1 m margin relative to the non-updated model, showcasing superior performance on all metrics against the finetune strategy. This approach is highly effective in maintaining the utility and accuracy of the model over time, avoiding retraining at regular intervals that consume significant computational resources, thus facilitating smarter and more responsive urban water management practices.

The methodology developed and implemented in this study lays the foundation for enhancing the resilience and efficiency of urban drainage networks. However, there remains scope for further investigation into various intricacies. One such avenue is the optimization of the model updating process. Currently, new data is integrated with each update cycle, but future research might explore prioritizing high-quality data samples to streamline this process. Central to this inquiry would be developing criteria for evaluating data sample quality and determining the triggers for updates.

CRedit authorship contribution statement

Siyi Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Jiaying Wang:** Writing – review & editing, Project administration, Data curation. **Kunlun Xin:** Supervision, Resources. **Hexiang Yan:** Supervision, Resources. **Shuping Li:** Supervision, Resources. **Tao Tao:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2024.122614.

Data availability

Data will be made available on request.

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